

EMBEDDED SYSTEMS LAB SETUP & SENSOR INTERFACING

Task – 2 Report

Motion Detection System using PIR Sensor and Arduino UNO

1. Objective

The objective of this project is to interface a PIR (Passive Infrared) motion sensor with an Arduino UNO microcontroller and develop a basic motion detection system. The system detects movement and automatically turns ON an LED when motion is detected.

This project helps in understanding:

- Embedded systems basics
- Sensor interfacing
- Digital input processing
- Real-time automation using Arduino

2. Introduction

Embedded systems are widely used in automation and smart electronic devices. Sensors play an important role in detecting environmental conditions and providing input data to microcontrollers.

In this project, a PIR motion sensor is used to detect human movement. The Arduino UNO continuously reads the sensor output and performs an action based on the received signal.

When motion is detected:

- The PIR sensor outputs a HIGH signal
- Arduino processes the signal
- LED turns ON
- Status message is displayed on Serial Monitor

When no motion is detected:

- LED remains OFF

This type of system is commonly used in:

- Home security systems
- Automatic lighting systems
- Smart surveillance systems
- Energy-saving automation

3. Componenets Required

S.No	Component Name	Quantity
1	Arduino UNO	1
2	PIR Motion Sensor	1
3	LED	1
4	220Ω Resistor	1
5	Breadboard	1
6	Jumper Wires	As required

4. Software Used

Software	Purpose
Tinkercad Circuits	Circuit simulation and coding
Arduino IDE (Built-in in Tinkercad)	Writing and executing Arduino code

5. Circuit Connections

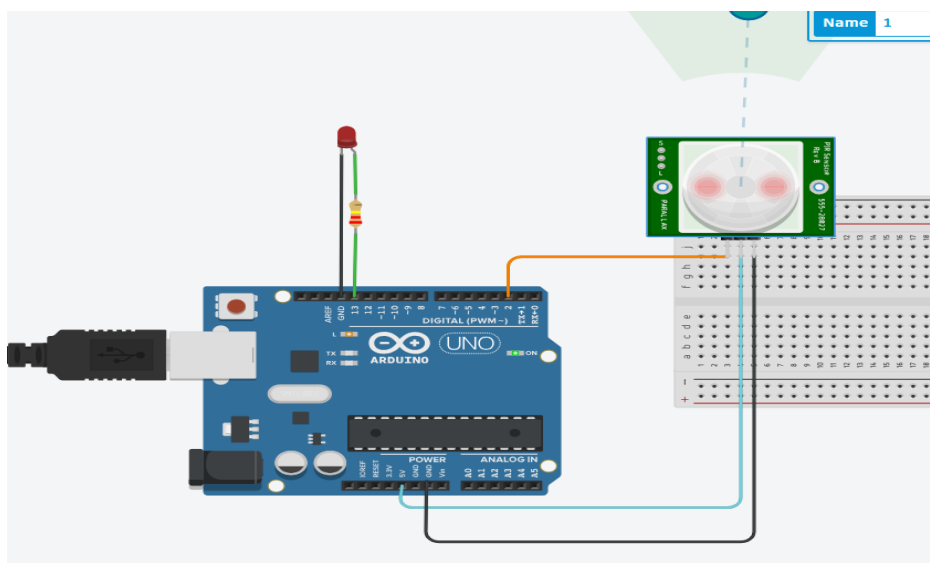
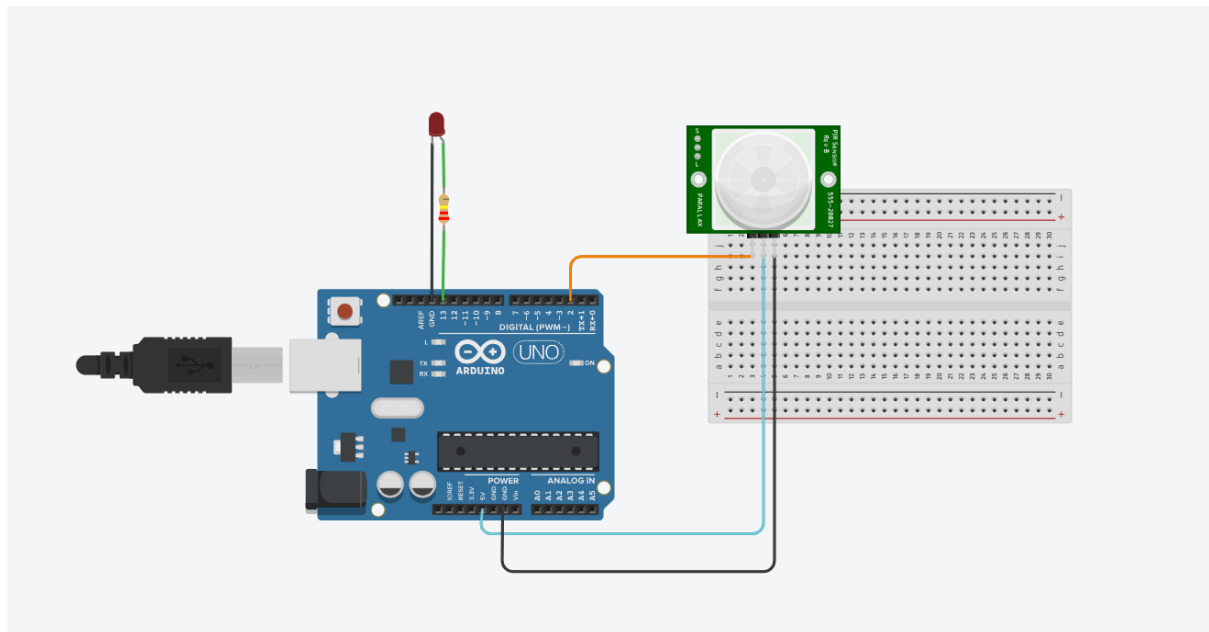
PIR Sensor Connections

PIR Pin	Arduino Connection
Signal	Digital Pin 2
Power	5V
Ground	GND

LED Connections

LED Terminal	Connection
Positive (Long Leg)	Pin 13 through 220 Ω resistor
Negative (Short Leg)	GND

6. Circuit Diagram



7. Arduino Program

```
int pirPin = 2;
```

```
int ledPin = 13;
```

```
void setup()
```

```
{
```

```
  pinMode(pirPin, INPUT);
```

```
  pinMode(ledPin, OUTPUT);
```

```
  Serial.begin(9600);
```

```
}
```

```
void loop()
```

```
{
```

```
  int motion = digitalRead(pirPin);
```

```
  if (motion == HIGH)
```

```
  {
```

```
    digitalWrite(ledPin, HIGH);
```

```
    Serial.println("Motion Detected!");
```

```
  }
```

```
  else
```

```
  {
```

```
    digitalWrite(ledPin, LOW);
```

```
    Serial.println("No Motion");
```

```
  }
```

```
  delay(500);
```

```
}
```

8. Working Principle

The PIR sensor detects infrared radiation emitted by humans or moving objects. Whenever motion occurs within the sensing range, the PIR sensor sends a HIGH signal to the Arduino UNO.

The Arduino continuously checks the sensor status using the `digitalRead()` function.

- If the sensor output is HIGH:
 - Motion is detected
 - LED turns ON
 - Message is printed on Serial Monitor
- If the sensor output is LOW:
 - No motion detected
 - LED turns OFF

The LED acts as an indicator for motion detection.

9. Output

When Motion is Detected

- LED turns ON
- Serial Monitor displays:

Motion Detected!

When No Motion is Detected

- LED turns OFF
- Serial Monitor displays:

No Motion

10. Applications

- Home Security Systems
- Automatic Door Systems
- Smart Lighting
- Motion Activated Alarms
- Energy Efficient Automation

11. Advantages

- Simple and low-cost system
- Easy to implement
- Energy efficient
- Real-time motion sensing
- Useful in IoT and automation projects

12. Conclusion

The PIR sensor was successfully interfaced with Arduino UNO using Tinkercad simulation. The system accurately detected motion and controlled the LED accordingly.

This project provided practical knowledge about:

- Sensor interfacing
- Embedded programming
- Real-time input/output control
- Arduino-based automation systems

The project demonstrates a basic but important application of embedded systems in security and smart automation.

13. Result

The motion detection system using PIR sensor and Arduino UNO was successfully designed, simulated, and tested in Tinkercad.

14. References

- [Arduino Official Website](#)
- [Tinkercad Circuits](#)
- [Arduino PIR Sensor Tutorial](#)